ETX5651

R scripts for ETX5651 assignment, analysis of quarterly GDP growth and financial conditions in the United States and Australia. The project combines layer 1 linear dependence analysis and layer 2 quantile matrix autoregressive connectedness analysis.

Repository contents

File or directory	Purpose
`Ass.Rproj`	RStudio project configuration
`layer1.R`	Data validation, descriptive statistics, stationarity diagnostics and self-normalized correlation tests
`layer2.R`	Quantile MAR estimation, generalized variance decompositions and two connectedness figures
`data.xlsx`	Analysis-ready quarterly data used by both scripts by default
`US_Australia_GDP_FCI_quarterly.xlsx`	Workbook with `Quarterly data`, `Quarterly inputs` and `Definitions` sheets
`results_layer1/`	Existing descriptive, stationarity and correlation outputs for layer 1
`results_layer2/`	Existing connectedness and quantile-asymmetry PDFs for layer 2

Data

`data.csv` contains **176 observations, from 1976Q4 to 2020Q3**, with consecutive, unique calendar quarters and no missing-value markers in the four analysis columns.

Column	Meaning	Treatment in the scripts
`quarter`	Calendar quarter, formatted as `YYYYQ1` through `YYYYQ4`	Used for ordering and continuity checks
`us_gdp_growth_qoq_pct`	US GDP growth, percent quarter on quarter	Used as supplied
`us_nfci`	US National Financial Conditions Index	Used in levels
`aus_gdp_growth_qoq_pct`	Australian GDP growth, percent quarter on quarter	Used as supplied
`aus_fci`	Australian Financial Conditions Index	Used in levels

GDP growth is already expressed in percentage units and is not annualized.

Provenance recorded in the workbook

The workbook's `Definitions` sheet records the following provenance.

- US GDP: BEA via FRED, series `GDPC1`.
- Australian GDP: IMF International Financial Statistics via FRED, series `NGDPRSAXDCAUQ`.

- US financial conditions: Chicago Fed NFCI via FRED, aggregated as the arithmetic mean of weekly observations whose Friday falls in each quarter.
- Australian financial conditions: Hartigan and Wright (2021), RBA Research Discussion Paper 2021-03, supplementary file `Data/fci_q_1_s_1_p_1_rdp.csv`.

Requirements

Install the packages in an R session:

```
install.packages(c("readxl", "ggplot2", "testcorr", "urca", "quantreg"))
```

Run the analysis

Open `Ass.Rproj` in RStudio, or use a terminal with the repository root as the working directory. The following examples use new output directory names to preserve the bundled results:

```
Rscript layer1.R data.csv results_layer1_new
Rscript layer2_qmar.R data.csv results_layer2_new
```

Without arguments, both scripts read `data.csv`; output defaults are `results_layer1` and `results_layer2`.

Layer 1: correlation diagnostics

`layer1.R` stacks variables as `(US_GDP, US_NFCI, AUS_GDP, AUS_FCI)` and performs:

- Quarter validation, sorting, numeric/finite-value checks and constant-series checks.
- Descriptive statistics and median/MAD-based outlier flags; flagged observations are retained.
- ADF tests with an intercept and BIC lag selection up to eight quarters, plus KPSS level-stationarity tests with truncation lag eight.
- Self-normalized ACF tests for all four series and CCF tests for all six pairs in both directions, using `testcorr`.
- Individual-lag inference, with lags 1 and 8 highlighted, and cumulative directional tests over lags 1–8. The significance level is 5% and `lambda = 0`.

Root outputs include `stacked_data.csv`, summary statistics, missing/outlier reports, the metadata template, stationarity diagnostics, four ADF text summaries and `series.pdf`. The `full_sample/` subdirectory contains the correlation matrix, individual and cumulative test tables, `ACF_SN.pdf`, `CCF_both_directions_SN.pdf` and `analysis.rds`.

Layer 2: quantile connectedness

`layer2_qmar.R` uses the order `(US_GDP, AUS_GDP, US_NFCI, AUS_FCI)`, corresponding to column stacking of a country-by-variable matrix.

It writes two files under the selected output directory:

Output	Interpretation
`figures/connectedness_heatmap.pdf`	Eight-quarter variance shares at each quantile; rows are recipients, columns are source innovations, and each row sums to 100%.
`figures/quantile_asymmetry.pdf`	Lower-quantile minus median financial-to-growth share over horizons 1–20 quarters, in percentage points.